DR. ISHRAT-UL-RBAD KHAN INSTITUTE OF ORAL HEALTH SCIENCES
DOW UNIVERSITY OF HEALTH SCIENCES
ANATOMY-BCQ CLASS TEST #2

Name:
Roll no.:
Date: 18/05/24

Max. Marks: 32
Min. Marks: 16

Each BCQ carries one (01) mark. There will be no negative marking

1. Primary follicles grow mainly under the influence of:
a) Estrogen
b) Follicle-stimulating hormone (FSH)
c) Luteinizing hormone (LH)
d) Progesterone
e) Testosterone

2. Mucosa of the uterus at the time of implantation is normally in which phase of menstrual cycle?
a) Follicular phase
b) Menstrual phase
c) Ovulatory phase
d) Proliferative phase
e) Secretory phase

3. A spermatozoon must be capacitated to penetrate through which barrier of the oocyte?
a) Corona radiata
b) Oocyte cell membrane
c) Oocyte nuclear membrane
d) Perivitelline space
e) Zona pellucida

4. Which of the following is an important outcome (result) of fertilization?
a) Chromosomal determination of sex of the embryo.
b) Completion of first meiotic division.
c) Completion of second meiotic division.
d) Formation of male and female pronuclei.
e) Formation of the second polar body

5. In the second week of development, the extra embryonic mesoderm disappears forming the:
a) Amniotic cavity
b) Chorionic cavity
c) Exocoelomic cavity
d) Secondary yolk sac
e) Syncytial lacunae

6. Which of the following events does not occur during first week of development?
a) Bilaminar germ disc formation
b) Blastocyst formation
c) Cleavage
d) Fertilization
e) Ovulation

7. **A trophoblastic villous characterized by the outer syncytial layer, inner cytotrophoblast and the mesoderm core differentiating into blood vessels, is known as:**

a) Free villous
b) Primary villous
c) Secondary villous
d) Stem villous
e) Tertiary villous

8. **Lab. Report of an expecting lady reveals high levels of human chorionic gonadotropin (HCG), and her fetal scan shows placental/trophoblastic tissue but no evidence of embryo. What is the likely diagnosis?**

a) Eeclampsia
b) Ectopic pregnancy
c) Hydatidiform mole
d) Sacrococcygeal teratoma
e) Sirenomelia

9. **Regarding primary yolk sac (umbilical vesicle), choose the wrong statement:**

a) It accommodates the primordial germ cells.
b) It acts as an early haemopoeitic organ.
c) It is bounded by the exocoelomic membrane.
d) It provides nourishment to the embryo.
e) The epiblast forms its roof.

10. **The most common site for the ectopic pregnancy is:**

a) Ampulla of fallopian tube
b) Cervix
c) Isthmus of fallopian tube
d) Ovary
e) Pouch of Douglas

11. **Select the correct order of structures representing the steps of formation of the notochord:**

a) Notochordal canal, notochordal plate, notochordal process, definitive notochord.
b) Notochordal plate, notochordal canal, definitive notochord.
c) Notochordal plate, notochordal process, notochordal canal, definitive notochord.
d) Notochordal process, notochordal canal, notochordal plate, definitive notochord.
e) Notochordal process, notochordal canal, definitive notochord.

12. **Normal fate of the notochord is:**

a) Intervertebral disc
b) Spinal cord
c) Nucleus pulposus
d) Vertebral arch only
e) Vertebral column

13. **Regarding the process of neurulation, choose the odd statement from the following:**

a) Cranial neuropore normally closes at 25th day of development.
b) Failure of the neural tube closure in the cranial part, results in spina bifida.
c) Neural tube is the primordium of central nervous system.
d) Neurulation is the process of formation of neural tube.

e) Notochord induces the overlying ectoderm to form neural plate.

14. Which of the following is a derivative of ectoderm?
a) Blood vessels
b) C-cells of thyroid gland
c) Central nervous system
d) Epithelial lining of the respiratory tract
e) Epithelial lining of the urinary bladder

15. Kidneys and gonads develop from which of the following germ layers?
a) Ectoderm
b) Endoderm
c) Intermediate mesoderm
d) Lateral plate mesoderm
e) Paraxial mesoderm

16. Pertaining to somites, which of the following statements is false/incorrect?
a) They are the segments of paraxial mesoderm.
b) Their first pair appears at about 20th day of development.
c) Their first pair appears in the cervical region.
d) They are used to determine the age of an embryo
e) Their dermatome component forms dermis of the back

17. The movements of the fetus are felt by the mother during which month?
a) 4th month
b) 5th month
c) 6th month
d) 7th month
e) 8th month

18. Contents of the umbilical cord include:
a) One artery, one vein and Wharton jelly.
b) One artery, two veins and Wharton jelly.
c) Only two arteries and Wharton jelly.
d) Two arteries, one vein and Wharton jelly.
e) Two arteries, two veins and Wharton jelly.

19. Normal volume of amniotic fluid at 37 weeks of gestation is approximately:
a) 400-600 ml
b) 600-700 ml
c) 800-1000 ml
d) 1500-2000 ml
e) >2000 ml

20. Mark the odd/false statement regarding human placenta:
a) It is of hemochorial type.
b) Its fetal portion is formed by chorionic frondosum.
c) Its maternal portion is formed by decidua parietalis.
d) Its maternal portion is characterized by 15-20 cotyledons.
e) It normally weighs about 500-600 gms.

21. **Peg and socket junction between the tooth and its socket is an example of:**
a) Gomphosis
b) Suture
c) Symphysis
d) Synchondrosis
e) Syndesmosis

22. **Regarding the following types of synovial joints with their examples, mark the wrong pair:**
a) Ball and socket--------Shoulder joint
b) Condylar----------Knee joint
c) Ellipsoid----------Ankle joint
d) Hinge----------Elbow joint
e) Plane synovial--------Intercarpal joint

23. **Actin filaments are present in which of the following connective tissue cell types?**
a) Adipocyte
b) Fibroblast
c) Histiocyte
d) Mast cell
e) Myofibroblast

24. **Signet ring appearance is the characteristic feature of:**
a) Adipocyte
b) Fibroblast
c) Histiocyte
d) Mast cell
e) Myofibroblast

25. **Cytoplasmic granules of which cell exhibit the phenomenon of metachromasia?**
a) Adipocyte
b) Fibroblast
c) Histiocyte
d) Mast cell
e) Plasma cell

26. **Choose the statement regarding connective tissue fibers which is true:**
a) Collagen fibers are acidophilic.
b) Elastic fibers branch extensively to form networks.
c) Elastic fibers are argyrophilic.
d) Reticular fibers contain type I collagen.
e) Type-V collagen is found in the basement membrane.

27. **Periosteum of bone is an example of:**
a) Adipose tissue
b) Loose areolar connective tissue
c) Dense irregular connective tissue
d) Dense regular connective tissue
e) Reticular connective tissue

28. **Which of the following is not a structural component of an osteon?**
a) Capillaries and nerve fibers

a) Diakinesis
b) Diplotene
c) Leptotene
d) Pachytene
e) Zygotene

Q26) Primary oocyte with zona pellucida surrounded only by simple or stratified layer of follicular/epithelial cells is appropriately termed as:

a) Antral follicle
b) Graafian follicle
c) Oogonium
d) Primary follicle
e) Primordial follicle

Q27) Regarding spermatogenesis, mark the odd/false statement:

a) Primary spermatocytes contain haploid number of chromosomes.
b) The developing spermatozoa are supported by sertoli cells.
c) Type-A pale spermatogonia give rise to type-B spermatogonia.
d) Type-B spermatogonia differentiate into primary spermatocytes.
e) Time required for spermatogenesis to complete is about 74 days.

Q28) The morphological changes that occur during spermiogenesis do not include:

a) Completion of meiosis-1.
b) Condensation of nucleus.
c) Formation of acrosome.
d) Formation of neck, middle piece and tail.
e) Shedding of most of the cytoplasm.

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